The two-row endpoint of the simultaneous-swap argument

Human Compression project: bounded L044 proof component

We use the complete accepted block-deficit and small-endpoint proofs, including their L047/Whiston hypotheses and dependencies. Thus $G \le A_n$ is a transitive string C-group with every maximal omission intransitive, a unique eligible i-edge joins the two G_i-orbits, and each of those orbits has two blocks. Put $s = \rho_i$. Products below act from left to right.

Lemma 1. *If one block size is two, $n \ge 12$, and $\alpha = 1$, then $r \le \lfloor (n-1)/2 \rfloor$. Consequently, together with the accepted small-endpoint proof, the simultaneous-swap case $m_1 = m_2 = 2$, $\alpha = 1$ is completely excluded as an obstruction to this rank bound.*

Proof. Write the two block sizes as $2, b$, where $b \ge 4$. A failure gives $\beta = \gamma = 0$, $r = b + 2$, and a single simultaneous fixed-point-free swap $\tau = \rho_{i-1}$ after reversal. Its evenness forces b even. The complete block image of G_i is the diagonal C_2 generated by τ.

Let $x = \rho_{i-2}$ if it exists, and let C be all other distinguished generators of G_i apart from τ, x. They centralize τ. As in the accepted proof, row projection of $\langle C \rangle$ is faithful: its kernel is contained in $\langle \tau \rangle$ and IP makes this intersection trivial. The group $\langle C \rangle$ is a character graph over a subgroup of $S_2 \times S_b$. Except for $c = \rho_{i+1}$, its generators fix both row anchors and have trivial block character. The independent-length bound in $S_1 \times S_{b-1}$ is $b - 2$. It forces both x and c to exist, gives $|C| = b - 1$, and identifies

$$H_0 = \langle C \setminus \{c\} \rangle = S_{b-1}$$

diagonally on the large orbit's nonanchor rows, fixing all four points of the small orbit.

The generators of H_0 split into the prefix at indices at most $i - 3$ and the tail at indices at least $i + 2$. These subgroups commute and have trivial intersection by IP. Since the natural S_{b-1} is centerless and directly indecomposable for $b - 1 \ge 3$, exactly one interval is nonempty. If the prefix is empty, x centralizes all of H_0. It preserves each large block: it fixes its crossing endpoint and its block image is either the identity on both systems or the simultaneous exchange. The natural S_{b-1} has one fixed point in each b-point block and trivial centralizer there, so x fixes the entire large orbit. On the small orbit it fixes the two-point block containing its crossing endpoint, and can perform at most one transposition on the other block. It is therefore the identity or odd, impossible.

Hence all of H_0 is the left prefix, and c is the sole right-interval generator. It centralizes H_0, so its large row projection is trivial. Faithfulness of row projection forces its small row projection to be the transposition; otherwise c would be the identity. Thus the row group is $S_2 \times S_{b-1}$, and the character of c is the only remaining possible nontrivial block character. If that character is nontrivial, c exchanges the two large blocks row by row. Since x commutes with c and fixes the large crossing endpoint, it also fixes its opposite row point. The prefix fixes both, and τ, c only exchange them. Their pair is then a G_i-invariant proper subset of the large orbit, a contradiction. Hence this character is trivial.

Label the small orbit A_0, A_1, A_0', A_1' and the large blocks $0, 1, \dots, b-1$ and $0', 1', \dots, (b-1)'$, with crossing endpoints $A_0, 0$. We have

$$c = (A_0\ A_1)(A_0'\ A_1'), \qquad \tau = (A_0\ A_0')(A_1\ A_1') \prod_{j=0}^{b-1} (j\ j'). \tag{1}$$

The prefix is a maximal independent string of S_{b-1}, and x centralizes its final-end omission.

First take even $b \geq 6$. The accepted L047 classification makes this prefix an ordinary Coxeter string. Its exceptional outer degree-six action does not occur, since $b-1$ is odd. After relabeling, let its final generator be

$$p = (b-2\ b-1)((b-2)'\ (b-1)').$$

The final-end omission is S_{b-2} on rows $1, \ldots, b-2$, fixing $0, b-1$. Thus x is the identity on the first large block, since it fixes 0, and its second-block action is either the identity or $(0'\ (b-1)')$. The latter is necessary to connect the large anchor pair to the other rows. On the small orbit x can only swap A'_0, A'_1. Evenness now forces

$$x = (A'_0\ A'_1)(0'\ (b-1)'). \tag{2}$$

These literal permutations satisfy

$$c = \left[\tau(xp)^2 x\right]^2 p. \tag{3}$$

Indeed, putting $u = b-2$ and $v = b-1$, the bracketed word has cycles

$$(A_0\ A'_1\ A_1\ A'_0), \qquad (0\ 0'), \qquad (u\ v'\ v\ u')$$

on these ten points. Its square is cp. On every other row pair it is just τ, whose square is the identity. Equation (3) makes the distinguished generator c redundant.

It remains to treat $b = 4$ explicitly. The prefix is

$$p = (1\ 2)(1'\ 2'), \qquad q = (2\ 3)(2'\ 3'),$$

in that order before x, τ, s, c. The centralizer of p on each four-point block gives

$$x = (A'_0\ A'_1)^\eta (1\ 2)^\epsilon (1'\ 2')^\delta (0'\ 3'), \qquad \eta + \epsilon + \delta = 1 \pmod 2. \tag{4}$$

The factor $(0'\ 3')$ is necessary for the large orbit's transitivity, and fixation of the two crossing endpoints excludes every other factor. If $\epsilon = \delta$, then $\eta = 1$ and $x_0 = xp^\epsilon = (A'_0\ A'_1)(0'\ 3')$. The same ten-point calculation as (3), now using the final prefix generator q, gives

$$c = \left[\tau(x_0 q)^2 x_0\right]^2 q. \tag{5}$$

The word uses only distinguished generators other than c, so both of these cases are excluded.

If $\epsilon \neq \delta$, then $\eta = 0$ and x fixes the small orbit pointwise. The subgroup $\langle p, q, x\rangle$ has on the large orbit the three point orbits $\{0\}$, $\{1,2,3\}$ and $\{0', 1', 2', 3'\}$, with natural symmetric projections on the last two. As s centralizes this subgroup, it fixes those seven points individually. Its crossing action and evenness therefore give

$$s = (A_0\ 0)h, \qquad h \in \{(A_1\ A'_0), (A_1\ A'_1), (A'_0\ A'_1)\}. \tag{6}$$

Put $w = s(\tau s x)^5 \tau x$. If $h = (A_1\ A'_1)$, the ten-point identity of the accepted four-row endpoint gives $w = p$. For the other two choices one has

$$p = (\tau s \tau w)^3. \tag{7}$$

For a direct check, $\tau s \tau w$ is the product of p and the following two disjoint three-cycles:

h	three-cycles
$(A_1\ A'_0)$	$(A_0\ 0\ 3)$, $(A_1\ A'_0\ 3')$,
$(A'_0\ A'_1)$	$(A_0\ 0\ 3)$, $(A'_0\ 3'\ A'_1)$.

These formulas hold for both choices (ϵ, δ). Cubing gives (7). Thus all six remaining choices make p redundant. This completes the case $b = 4$ and the proof. □

Scope and attribution. This is a further bounded proof of the both-imprimitive rank branch in L044, Section 5.1, physical pages 23–25. It retains the exact common-theory and source attribution of the accepted block-deficit note. The explicit word identities above are new proofs for the residual endpoint, with complete cycle verification; bounded word-finding experiments are not required premises. Together with the accepted endpoint note, they complete the $\alpha = 1$ two-block case. They do not settle $\alpha = 0$, mixed block-count configurations, or all of L044. There is no new original or conjecture credit and no new formal declaration.